

Every screen you will see, before you see it.

This is the SPLITSTEEL sitting on paper: the set-up, then every screen the kit shows in the order it shows them, then the objects it leaves behind and how the halves are placed for your heirs. The screens are not drawings. They were recorded by driving the real program, build **0396da41**, through a complete sitting; the website replays the same frames, and the test suite regenerates them so they cannot drift from the code.

Before you sit down

On the table

- The Raspberry Pi with the SPLITSTEEL card, and its power supply
- Any HDMI screen and a USB keyboard
- Two casino dice, two colours, so first and second stay distinct
- Two USB sticks, any size; they become **R** and **B**
- The three printed cards: dice card, ARITHMETIC CARD, SEED GRID CARD (field manual pages 11 to 13; the boxed kit ships them printed)
- Pencil and paper

Not in the room

- No network lead and no Wi-Fi hardware on the machine
- No phone on the table, no camera that can see the screen
- Nobody who does not need to be there

For afterwards

- Steel plates, the punch jig and letter punches, or an engraver
- Two M-DISCs and a drive that burns them
- Two ordinary computers, one for each half
- Two hiding places in two buildings

The rules

- **Practice first.** [3] at the menu is the whole sitting with toy values, nothing at stake.
- **Either half alone is nothing.** Both halves together are the key.
- **The two halves never meet** on one computer or in one hiding place.
- **Power off with [p]**, never by pulling the plug. Wait for the green light to stop.
- **Never type a seed into a website.** Not this one, not any.

The seed in these screens is a published demonstration value: **critic next cloud athlete balcony timber gasp draft paddle silk doll glass**. It must never hold money.

Power on

Power on, and the machine checks itself before it lets you in.

```

  _____
 /  _  \  /  _  \  /  _  \  /  _  \  /  _  \
|  _  | |  _  | |  _  | |  _  | |  _  |
|  _  | |  _  | |  _  | |  _  | |  _  |
|  _  | |  _  | |  _  | |  _  | |  _  |
 \__  \ \__  \ \__  \ \__  \ \__  \
  _____

```

the Pi is a blender - steel is the vault

Starting up. This can take a few minutes - the machine is small and it checks itself before it lets you in. Leave it be.

Nothing on this computer talks to anything: no wi-fi, no bluetooth, no network. Nothing you type is ever written to the card. When the power goes off, everything it knew is gone.

The menu will appear by itself. There is nothing to press.

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MAIN MENU

integrity: wordlist verified - 10 official vectors pass
wizard build 0396da41 boot: quiet reached the menu 41 s after power-on

[1] build a vault	words -> pad + ciphertext
[2] recover	pad + ciphertext -> words
[3] practice	the whole sitting, toy values
[?] help	new to this? start here
[p] power off	the safe way to finish - and the ONLY safe way to remove the card. Wait for the light.

First time with this kit? Do [3] practice before anything real: the whole sitting with toy values, and nothing on the table that matters.

choose:1

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BEFORE ANYTHING

This machine must be a blender: no network lead, no Wi-Fi hardware, no camera that can see this screen, no phone on the table.

Nothing you type is ever written to disk. Power off ends everything.
Is the room clean and the machine air-gapped? [y/n]y

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THE USB STICKS

Each half or part gets its own USB stick, written one at a time so that no two of them can ever end up on the same one.

3 removable drives are plugged in. Good.

```

/dev/sda1 (KINGSTON, 15.5 GB)
/dev/sdb1 (SANDISK, 31.0 GB)
/dev/sdc1 (PNY, 7.7 GB)

```

Leave them in for now. When it is time to write, I will ask you to unplug all but one.
press enter to continue - the screen will then clear

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BUILD THE VAULT - THE SEED

Where do the words come from?

[1] I already have a seed	type in the words I have
[2] make a brand new one	if the seed I have might be weak or seen

choose:2

SCREEN 01

Power on. The Pi checks itself for a minute or two and the menu appears by itself. Nothing on this machine talks to anything, and nothing typed is ever written to the card.

SCREEN 02

The menu. The build code under the title is the hash of the program itself; the manual and the website tell you what it should read. On a kit there is no shell to quit to, only [p] to power off.

You type 1 and press enter.

SCREEN 03

The air-gap question. The Pi has no radios and no network lead, but the room matters too: no camera on the screen, no phone on the table.

You type y and press enter.

SCREEN 04

Before a single die is rolled the wizard counts the sticks, so nobody finds out forty minutes in that they brought one.

You press enter.

SCREEN 05

Two ways in: type the words you already have, or mint a brand new seed. This sitting mints one.

You type 2 and press enter.

A brand new seed

Mint a seed from your dice and this machine together, and check the arithmetic by hand on the cards.

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A BRAND NEW SEED

This makes a wallet that has never existed before. It is for someone whose current seed might be weak or might have been seen - a device with a bad random number generator, a phrase typed into a website, a backup that was photographed.

This is a bigger thing than backing up a seed you already have. Everything you own will depend on the next few minutes. The arithmetic here is standard and tested against the official vectors, but no outside party has reviewed this program.

Make a new seed? [y/n]y

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A BRAND NEW SEED - HOW LONG

[1] 12 words 128 bits. Ample. About 36 dice rolls.
 [2] 24 words 256 bits. Twice the rolls, same method.
 choose [enter = 12]:

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A BRAND NEW SEED - WHERE THE RANDOMNESS COMES FROM

- [1] **your dice AND this machine, mixed** **recommended**
 About 36 rolls. The machine commits to its number first, on screen, before you roll - so it cannot steer the result. The two are combined, and the answer is good if EITHER source is good. Safe if this card was tampered with. Safe if your dice are loaded. Same effort as dice alone.
 You still trust that the words shown are the ones the numbers make - which you check by hand, afterwards.
- [2] **your dice only**
 About 36 rolls. Trusts nothing but your dice being fair, and your arithmetic. Nothing else at all.
- [3] **this machine only** **quick**
 Instant, no dice. Three sources mixed: the kernel's pool (which hands out nothing before it has gathered enough entropy), the board's own hardware noise, and the timing of your answers. Weak only if all three fail at once. What you cannot check from outside is that whoever flashed this card did not swap the program: the card is not reproducible from source byte for byte, so there is no published number to check it against. Still far better than keeping a seed you already doubt.

A seed invented by a hardware wallet is not one of these three. That is the whole reason this step exists: make a new one here, then move your money to it.

choose:1

SCREEN 06

Minting a wallet that has never existed is a bigger thing than backing one up, and the wizard says so before it starts.

You type **y** and press enter.

SCREEN 07

Twelve words is 128 bits and ample. Twenty-four is the same method with twice the rolls.

You press enter.

SCREEN 08

Dice and machine mixed is the recommendation: the result is good if either source was good, so a tampered card or loaded dice each fail alone.

You type **1** and press enter.

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A BRAND NEW SEED - THE MACHINE COMMITS FIRST

This is the machine's half. It is fixed now, before you touch a die.
Mixed by exclusive-or from three sources: the kernel's random pool,
the board's own hardware noise read directly, and the timing of your
answers so far. Good if any one of the three is good.

THE MACHINE'S NUMBER - write it in grid A of the SEED GRID CARD, box for box

1	2	3	4	
6DCE	ACDC	3EB3	F515	
5	6	7	8	check
6F05	65C2	F485	0AAE	FE58

Write this down. Later you can check the arithmetic yourself,
on the printed table, and prove this machine did not
choose your seed for you.
press enter when it is written down - then roll

SCREEN 09

The machine's half is drawn and
shown before any die is rolled, so it
cannot steer the result after seeing
your dice. Copy it into grid A of the
SEED GRID CARD, box for box.

You press enter.

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A BRAND NEW SEED - NOW YOUR DICE

The dice half is cast.
Write it in grid B of the SEED GRID CARD, box for box, check group too.

1	2	3	4	
5E1C	0473	B9A2	E8D1	
5	6	7	8	
370F	6A5C	2D94	081F	
check				
73C4				

press enter to continue - the screen will then clear

SCREEN 10

Your half is cast, with its check
group. Copy it into grid B of the
SEED GRID CARD.

You press enter.

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A BRAND NEW SEED - MIXED

A combined with B, exclusive-or, gives C - THE NEW SEED, AS HEX.
Good if either source was good. Both would have to be bad, at
the same time, for this to be weak.

A the machine's number					B your dice				
1	2	3	4		1	2	3	4	
6DCE	ACDC	3EB3	F515		5E1C	0473	B9A2	E8D1	
5	6	7	8	check	5	6	7	8	check
6F05	65C2	F485	0AAE	FE58	370F	6A5C	2D94	081F	73C4
C the new seed = A combined with B									
1	2	3	4						
33D2	A8AF	8711	1DC4						
5	6	7	8	check					
580A	0F9E	D911	02B1	8D9C					

Write all three grids onto the SEED GRID CARD, box for box.
Grids A and B should already be there. With all three on paper
you can prove the arithmetic yourself, next.
press enter to continue - the screen will then clear

SCREEN 11

A combined with B gives C, the
new seed. Both would have to be
bad, at the same time, for this to be
weak. All three grids go onto the
card.

You press enter.

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A BRAND NEW SEED - CHECK THE ARITHMETIC

You have three numbers on paper: the machine's, your dice, and the seed. The seed should be the first two combined with exclusive-or.

Checking that yourself is what takes this program off the trust path. Until you do, you are taking its word that it used the number it showed you.

You will need two cards in your hand:

the SEED GRID CARD grids A, B and C, written box for box
the ARITHMETIC CARD the 16 x 16 table you read them against

Both are in the kit box, in the field manual (pages 12 and 13) and on the website. Use the printed table, never a screen:
a table this machine drew is a table it could bend.

[1] I have both - check every group now 8 lookups, about four minutes

[2] I want to get them - I will check later from my paper

choose:1

SCREEN 12

Now you check the machine, not the other way round. It asks for the SEED GRID CARD you have filled in and the printed ARITHMETIC CARD, and says where to find them.

You type **1** and press enter.

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A BRAND NEW SEED - CHECK THE ARITHMETIC

The lit box in A is the row on the table, the lit box in B the column; where they meet is that box of C. Type its four characters. Move freely: a box number jumps there, b goes back, v checks all 8.

A the machine's number

1 2 3 4

6DCE ACDC 3EB3 F515

5 6 7 8

6F05 65C2 F485 0AAE

FE58

check

B your dice

1 2 3 4

5E1C 0473 B9A2 E8D1

5 6 7 8

370F 6A5C 2D94 081F

73C4

check

C the new seed = A combined with B

1 2 3 4

???? ???? ???? ????

5 6 7 8

???? ???? ???? ????

8D9C

check

where A1 and B1 meet:33d2

SCREEN 13

For each lit box: the A character is the row on the printed table, the B character the column, and where they meet is C. Box 4 is mistyped here on purpose: it goes red and is retyped.

You type **33d2** and press enter.

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A BRAND NEW SEED - THE ARITHMETIC

All 8 groups agree.

A the machine's number

1 2 3 4

6DCE ACDC 3EB3 F515

5 6 7 8

6F05 65C2 F485 0AAE

FE58

check

B your dice

1 2 3 4

5E1C 0473 B9A2 E8D1

5 6 7 8

370F 6A5C 2D94 081F

73C4

check

C the new seed = A combined with B

1 2 3 4

33D2 A8AF 8711 1DC4

5 6 7 8

580A 0F9E D911 02B1

8D9C

check

You have now checked, by hand, that the seed really is the machine's number combined with your dice. This program did not choose your wallet for you, and you did not have to take its word for it.

press enter to continue - the screen will then clear

SCREEN 14

All eight groups agree. The program did not choose your seed, and you did not have to take its word for it.

You press enter.

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A BRAND NEW SEED - THE WORDS

from your dice mixed with this machine

1 critic

5 balcony

9 paddle

2 next

6 timber

10 silk

3 cloud

7 gasp

11 doll

4 athlete

8 draft

12 glass

Write these down now, on paper, in order.

The last word is not free - part of it is a checksum over all the others, which is how a wallet knows the phrase is whole. Copy it exactly like the rest.

press enter when every word is written down

SCREEN 15

The seed as its twelve words. These are written on paper and read back in the next screen, the same way every entry on this machine is read back.

You press enter.

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A BRAND NEW SEED - READ THEM BACK

Enter the 12 words, in order.

A unique start is enough - 'lega' becomes 'legal'.

Move about freely - nothing you have typed is lost by moving.

Type a command or the characters, then press enter:

1 to 12

b

enter

q

go straight to that word

back one word, leaving it as it is

keep this word and move on

give up on this entry

1

2

3

4

5

6

7

8

9

10

11

12

word 1 of 12:critic

SCREEN 16

The words typed back from your paper, in the same grid as everything else. The checksum in the last word catches a slip.

You type **critic** and press enter.

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A BRAND NEW SEED - READ THEM BACK

These 12 words are a valid seed.

1

2

3

4

critic

next

cloud

athlete

5

6

7

8

balcony

timber

gasp

draft

9

10

11

12

paddle

silk

doll

glass

The checksum built into the last word agrees with the rest.

press enter to continue - the screen will then clear

SCREEN 17

Twelve words, read back exactly. The paper is right.

You press enter.

SPLITSTEEL · the Pi is a blender · steel is the vault

A BRAND NEW SEED - BEFORE YOU MOVE ANY MONEY

The words match your paper.

This seed has never touched a wallet.

Nothing has proved that any wallet will accept it, or that the addresses you are about to send to are ones you can spend from.

In this order:

- 1 Finish this sitting, so the new seed is split and backed up.
- 2 Restore the words into your wallet or hardware signer, **typing from your paper**, not from this screen.
The address you fund must come from the words you actually wrote down and read back - not from anything still in this machine's memory.
- 3 Send a **small test amount** and spend it back out again.
- 4 Only then move everything across.
- 5 Then retire the old seed: destroy its backups, so nobody in future recovers an empty wallet and thinks they have failed.

Do not skip 3. It is the only step that proves the words work.
press enter to continue - the screen will then clear

SCREEN 18

A new wallet is tested before it holds anything: a small amount in and back out first, and the old seed retired only after everything has moved.

You press enter.

EPISODE 03

The pad and the ciphertext

Cast the pad from your dice; the wizard works out the ciphertext. Two halves, written down exactly.

SPLITSTEEL · the Pi is a blender · steel is the vault

BUILD THE VAULT - HOW MANY PARTS

The seed can be split two ways. Both are the same arithmetic on paper; they differ in what a lost object costs, and in who could combine what.

- [1] **two halves, both needed** 2 of 2 RED pad, BLUE ciphertext
shortest: 32 characters an object, 32 lookups to recover
nobody you give a half to can ever spend it, alone or together
lose every copy of one half and it is gone: keep three copies of each
- [2] **three parts, any two recover** 2 of 3 RED, BLUE, GREEN
lose one part outright and nothing is lost
one part each to three places, or three people
twice the writing: 64 characters an object, 64 lookups to recover
any two holders together can recover it - choose them like cosigners

choose [enter = 1, q = back to the menu]:1

SCREEN 19

One question before any die is rolled: two halves, both needed, or three parts, any two of which recover. The screen states the trade both ways, and the default is the simple one.

You type **1** and press enter.

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BUILD THE VAULT - CAST THE PAD

The pad is YOUR dice, never this machine. If you have already rolled it at the table with the printed card, choose hex entry.

- [1] enter dice rolls, the wizard maps them
[2] enter the pad as hex, already rolled
Rolled fresh for THIS seed. Never type in a number you have used for another wallet: two backups made with one pad give away both.
- [q] stop here - back to the menu, nothing written

choose:1

SCREEN 20

The pad is your dice, never the machine. It is the other random number, the one the seed hides behind.

You type **1** and press enter.

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CAST THE PAD - DICE

The pad is cast.

1234

2D94081F5E1C0473

5678

B9A2E8D1370F6A5C

check

73C4

press enter to continue - the screen will then clear

SCREEN 21

The pad is cast, with the check group the wizard works out from it.

You press enter.

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BUILD THE VAULT - WRITE THE PAD

THE RED PAD - half the secret, write it exactly, check group too

1234

2D94081F5E1C0473

5678

B9A2E8D1370F6A5C

check

73C4

This goes on the keeper plate and the spare. No labels, ever.
written down, with its check group? press enter

SCREEN 22

The RED half. Write it exactly, check group too. This goes on the keeper plate and its spare, and carries no label, ever.

You press enter.

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BUILD THE VAULT - THE CIPHERTEXT

THE BLUE CIPHERTEXT - the other half, write it exactly, check group too

1234

1E46A0B0D90D19B7

5678

E1A8E74FEE1E68ED

check

FE58

Now the SPOT CHECK. Everything so far has been done by this machine.
This is the one step that does not trust it - you do the arithmetic yourself, from the XOR table on your printed card, and it grades you.

Use the table on the CARD, not on any screen. A dishonest machine could show you a doctored table; a printed card cannot be changed.

Haven't got the card? It is pages 11 and 12 of the field manual - the page headed "Print this page and take it to the table", with the dice card, and the ARITHMETIC CARD with the 16x16 XOR table right after it. You will find the manual:

on any SPLITSTEEL stick or disc: manual/
on the website: the field manual download
or the recipe page, which prints the table on its own

Print it on another computer and come back. This machine deliberately will not show you the table - a table it drew is a table it could bend.
Do the spot check now? (strongly recommended) [y/n]y

SCREEN 23

The BLUE half: the seed combined with the pad. Either half on its own is a random number. Then the spot check, done from the printed card and never from a screen.

You type y and press enter.

EPISODE 04

The checks

The spot check proves the machine's arithmetic against your printed card. The round trip proves your paper.

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BUILD THE VAULT - SPOT CHECK 1 OF 3

Find the pair on the XOR table on your card: the PAD character gives the row, the CIPHERTEXT character gives the column. Read off where they meet, and type it in.

RED PAD

1	2	3	4
2D94	081F	5E1C	0473
5	6	7	8
B9A2	E8D1	370F	6A5C

check

BLUE CIPHERTEXT

1	2	3	4
1E46	A0B0	D90D	19B7
5	6	7	8
E1A8	E74F	EE1E	68ED

check

character 10 of 32 - the second character of the lit box, box 3

pad E

ciphertext 9

row E, column 9 gives:7

SCREEN 24

Three characters picked at random. The RED PAD character is the row, the BLUE CIPHERTEXT character the column, and you read where they meet off the printed table.

You type 7 and press enter.

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BUILD THE VAULT - SPOT CHECK RESULT

OK character 10 you said 7, this machine says 7

OK character 21 you said 0, this machine says 0

OK character 26 you said 9, this machine says 9

All three agree.

The arithmetic on this machine matches the arithmetic on your card.

You are no longer taking its word for anything.

press enter to continue - the screen will then clear

SCREEN 25

Your table and the machine agree, so the machine's arithmetic is the arithmetic on your card.

You press enter.

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READ IT BACK

re-enter the RED PAD, from your own writing

Four characters per box, in the order they appear on the object.

Move about freely - nothing you have typed is lost by moving.

Type a command or the characters, then press enter:

1 to 9 go straight to that box

b back one box, leaving it as it is

enter keep this box and move on

q give up on this entry

1

2

3

4

5

6

7

8

check

box 1 of 8:2d94

SCREEN 26

Each half goes in box by box from the paper. The check box at the end catches a misread character and says which column of the paper to look at.

You type 2d94 and press enter.

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BUILD THE VAULT - ROUND-TRIP RESULT

1 critic

2 next

3 cloud

4 athlete

5 balcony

6 timber

7 gasp

8 draft

9 paddle

10 silk

11 doll

12 glass

MATCHES the seed you started with.

And they match the paper seed in front of you, word for word? [y/n]y

SCREEN 27

The twelve words come back exactly. The paper is right.

You type y and press enter.

EPISODE 05

The sticks

Each half onto its own stick, one at a time, read back and labelled. Then what is left to do.

SPLITSTEEL · QUICK START never type a seed into a website

9

SPLITSTEEL · the Pi is a blender · steel is the vault

BUILD THE VAULT - WRITE THE RED STICK

Plug in **ONE** stick: the one that will carry the **PAD**.
Unplug every other USB stick first.

This is the rule that makes the whole scheme safe: with one stick in the machine, both halves cannot land on the same one.

- [1] it is plugged in, and it is the only one
- [2] skip the USB copy of this half

choose:1

SCREEN 28

One stick at a time, with every other stick unplugged, so both halves can never land on the same one. The half, the plate files and the recovery tools go on, then it is read back.

You type **1** and press enter.

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BUILD THE VAULT - THE RED STICK IS WRITTEN

Written and verified.

Every file was read back and matched its checksum, and the half was re-derived from the stick to confirm the check group.

Now, before you do anything else:

Write R on this stick in permanent marker.

R = pad. It is safe to label: a half on its own is nothing.

Then unplug it.

press enter when it is labelled and unplugged

SCREEN 29

Read back and verified. Label it R or B in marker: a half on its own is nothing, so the label is safe. Then unplug it before the other one goes in.

You press enter.

SPLITSTEEL · the Pi is a blender · steel is the vault

BUILD THE VAULT - WRITE THE BLUE STICK

Plug in **ONE** stick: the one that will carry the **CIPHERTEXT**.
Unplug every other USB stick first.

This is the rule that makes the whole scheme safe: with one stick in the machine, both halves cannot land on the same one.

- [1] it is plugged in, and it is the only one
- [2] skip the USB copy of this half

choose:1

SCREEN 30

One stick at a time, with every other stick unplugged, so both halves can never land on the same one. The half, the plate files and the recovery tools go on, then it is read back.

You type **1** and press enter.

SPLITSTEEL · the Pi is a blender · steel is the vault

BUILD THE VAULT - THE BLUE STICK IS WRITTEN

Written and verified.

Every file was read back and matched its checksum, and the half was re-derived from the stick to confirm the check group.

Now, before you do anything else:

Write B on this stick in permanent marker.

B = ciphertext. It is safe to label: a half on its own is nothing.

Then unplug it.

press enter when it is labelled and unplugged

SCREEN 31

Read back and verified. Label it R or B in marker: a half on its own is nothing, so the label is safe. Then unplug it before the other one goes in.

You press enter.

SPLITSTEEL · the Pi is a blender · steel is the vault

BUILD THE VAULT - THE SITTING IS DONE

On the table you now have the pad and the ciphertext, each with a check group. What remains, in order:

OPEN START-HERE.html ON EACH STICK.

It is a page that opens with a double-click on any computer and walks that half through to finished: burning the disc, making the plate, and where to put it. You do not have to remember any of what follows - it is all on that page, per half.

THE TWO STICKS MUST NEVER MEET.

Take **R** to one computer and **B** to a different one, and keep it that way. Either stick alone is nothing. Both on one computer is your whole seed on that computer.

- 1 punch or engrave the keeper plates: one pad, one ciphertext
- 2 make the spare pair on your chosen medium
- 3 run RECOVER on this wizard reading only off the finished objects
- 4 only then destroy the old paper seed
- 5 split the copies: no hiding place holds both halves
- 6 wipe or destroy this SD, power off, post the heirs' letter
- 7 send a small test amount to the NEW wallet and spend it back out, then move everything across
- 8 only then retire the OLD seed: destroy its plates, discs and paper, so nobody later recovers an empty wallet and thinks they have failed

When you are finished with this machine, choose **p** at the menu to power off. Do not pull the plug.
press enter to clear the screen

SCREEN 32

What is left is physical. START-HERE.html on each stick walks that half through to its plate and its disc. The two sticks never meet on one computer.

You press enter.

EPISODE 06

Power off

The only safe way to finish, and the only safe way to take the card out.

Powering off. Nothing was ever stored.

The screen will go blank in a moment. That is NOT the end.

RED light = the plug is in. It never goes off. Ignore it.
GREEN light = the card is busy. Watch this one.

Pull the plug only when the GREEN light has been off for a slow count of ten. On this board that is a good half-minute after the screen goes blank. Pulling it sooner is how the card gets damaged.

SCREEN 33

Power off from the menu, never by pulling the plug. The red light stays on; wait until the green one has been off for a slow count of ten.

EPISODE 07

Three parts, any two

The same wallet as three parts, any two of which recover: numbers 1 and 2 from you, number 3 worked out, six lookups, three sticks.

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BUILD THE VAULT - THE SEED

Where do the words come from?

[1] I already have a seed type in the words I have
[2] make a brand new one if the seed I have might be weak or seen
choose:1

SCREEN 34

Two ways in: type the words you already have, or mint a brand new seed. This sitting mints one.

You type **1** and press enter.

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BUILD THE VAULT - THE WORDS

Enter the 12 words, in order.

A unique start is enough - 'lega' becomes 'legal'.

Move about freely - nothing you have typed is lost by moving.

Type a command or the characters, then press enter:

- 1 to 12** go straight to that word
- b** back one word, leaving it as it is
- enter** keep this word and move on
- q** give up on this entry

1	2	3	4
5	6	7	8
9	10	11	12

word 1 of 12:critic

SCREEN 35

A seed you already have goes in word by word, in the same grid as everything else. A unique start is enough, and the checksum in the last word catches a slip.

You type **critic** and press enter.

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BUILD THE VAULT - HOW MANY PARTS

The seed can be split two ways. Both are the same arithmetic on paper; they differ in what a lost object costs, and in who could combine what.

- [1] **two halves, both needed** 2 of 2 **RED** pad, **BLUE** ciphertext
shortest: 32 characters an object, 32 lookups to recover
nobody you give a half to can ever spend it, alone or together
lose every copy of one half and it is gone: keep three copies of each
- [2] **three parts, any two recover** 2 of 3 **RED, BLUE, GREEN**
lose one part outright and nothing is lost
one part each to three places, or three people
twice the writing: 64 characters an object, 64 lookups to recover
any two holders together can recover it - choose them like cosigners

choose [enter = 1, q = back to the menu]:2

SCREEN 36

One question before any die is rolled: two halves, both needed, or three parts, any two of which recover. The screen states the trade both ways, and the default is the simple one.

You type **2** and press enter.

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BUILD THE VAULT - CAST NUMBER 1

Number 1 is YOUR dice, never this machine. If you have already rolled it at the table with the printed card, choose hex entry.

- [1] enter dice rolls, the wizard maps them
- [2] enter number 1 as hex, already rolled
Rolled fresh for THIS seed. Never type in a number you have used for another wallet: two backups made with one pad give away both.
- [q] stop here - back to the menu, nothing written

choose:1

SCREEN 37

Three parts: number 1 is your dice, exactly as the pad was. Number 2 comes next, and number 3 the wizard works out from the seed and the other two.

You type **1** and press enter.

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CAST NUMBER 1 - DICE

The number 1 is cast.

1	2	3	4
5	6	7	8

check

press enter to continue - the screen will then clear

SCREEN 38

Number 1 is cast, with its check group.

You press enter.

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BUILD THE VAULT - CAST NUMBER 2

Number 2 is YOUR dice, never this machine. If you have already rolled it at the table with the printed card, choose hex entry.

- [1] enter dice rolls, the wizard maps them
- [2] enter number 2 as hex, already rolled
- Rolled fresh for THIS seed. Never type in a number you have used for another wallet: two backups made with one pad give away both.
- [q] stop here - back to the menu, nothing written

choose:2

SCREEN 39

Number 2 can be rolled at the table beforehand and typed in as hex, which is what happens here.

You type **2** and press enter.

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TYPE IN NUMBER 2

number 2 you rolled at the table

Four characters per box, in the order you rolled them.

Move about freely - nothing you have typed is lost by moving.

Type a command or the characters, then press enter:

- 1 to 8** go straight to that box
- b** back one box, leaving it as it is
- enter** keep this box and move on
- q** give up on this entry

1	2	3	4

5	6	7	8

box 1 of 8:4d6f

SCREEN 40

Eight boxes from the paper. There is no check group yet; the wizard works it out.

You type **4d6f** and press enter.

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BUILD THE VAULT - THE THREE NUMBERS

Write all three down, each with its check group. Number 3 is the seed combined with 1 and 2, so all three combined give the seed back.

NUMBER 1				NUMBER 2			
1	2	3	4	1	2	3	4
9C1E	4A7B	02D5	F386	4D6F	81A3	C5E7	092B
5	6	7	8	5	6	7	8
E1B0	C94D	7A2F	6E53	4D6F	81A3	C5E7	092B
				check			
				1BB7			
NUMBER 3 = the seed combined with 1 and 2							
1	2	3	4				
E2A3	6377	4023	E769				
5	6	7	8				
F4D5	4770	66D9	65C9				
				check			
				962B			

Then they pair up into three parts:

RED numbers 1 and 2

BLUE numbers 2 and 3

GREEN numbers 3 and 1

Any two parts hold all three numbers. One part alone is two random numbers. No hiding place ever holds two different parts. written down, all three with their check groups? press enter

SCREEN 41

Number 3 is the seed combined with 1 and 2, so all three combined give the seed back. RED takes 1 and 2, BLUE 2 and 3, GREEN 3 and 1: any two parts hold all three numbers, and one part alone is two random numbers.

You press enter.

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BUILD THE VAULT - THE SPOT CHECK

Now the SPOT CHECK. Everything so far has been done by this machine. This is the one step that does not trust it - you do the arithmetic yourself, from the XOR table on your printed card, and it grades you.

Use the table on the CARD, not on any screen. A dishonest machine could show you a doctored table; a printed card cannot be changed.

Haven't got the card? It is pages 11 and 12 of the field manual - the page headed "Print this page and take it to the table", with the dice card, and the ARITHMETIC CARD with the 16x16 XOR table right after it. You will find the manual:

on any SPLITSTEEL stick or disc: manual/
on the website: the field manual download
or the recipe page, which prints the table on its own

Print it on another computer and come back. This machine deliberately will not show you the table - a table it drew is a table it could bend.
Do the spot check now? (strongly recommended) [y/n]y

SCREEN 42

The same spot check as before, from the printed card and never from a screen.

You type y and press enter.

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BUILD THE VAULT - SPOT CHECK 1 OF 3

Two lookups on the XOR table on your card: number 1 meets number 2, then what you read off meets number 3. Type each one in turn.

NUMBER 1					NUMBER 2				
1	2	3	4		1	2	3	4	
9C1E	4A7B	02D5	F386		4D6F	81A3	C5E7	092B	
5	6	7	8	check	5	6	7	8	check
E1B0	C94D	7A2F	6E53	1BB7	4D6F	81A3	C5E7	092B	0000

NUMBER 3 = the seed combined with 1 and 2

1	2	3	4	
E2A3	6377	4023	E769	
5	6	7	8	check
F4D5	4770	66D9	65C9	962B

character 10 of 32 - the second character of the lit box, box 3
number 1 2
number 2 5
number 3 0
row 2, column 5 gives:7

SCREEN 43

Two lookups per character: number 1 meets number 2 on the printed table, then what you read off meets number 3. Six lookups for three characters.

You type 7 and press enter.

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BUILD THE VAULT - SPOT CHECK RESULT

OK character 10 you said 7, this machine says 7
OK character 10 you said 7, this machine says 7
OK character 21 you said 4, this machine says 4
OK character 21 you said 0, this machine says 0
OK character 26 you said F, this machine says F
OK character 26 you said 9, this machine says 9

All six agree.

The arithmetic on this machine matches the arithmetic on your card.
You are no longer taking its word for anything.
press enter to continue - the screen will then clear

SCREEN 44

Your table and the machine agree, so the machine's arithmetic is the arithmetic on your card.

You press enter.

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READ IT BACK

re-enter NUMBER 1, from your own writing

Four characters per box, in the order they appear on the object.
Move about freely - nothing you have typed is lost by moving.

Type a command or the characters, then press enter:

1 to 9 go straight to that box
b back one box, leaving it as it is
enter keep this box and move on
q give up on this entry

1 2 3 4


5 6 7 8


check



box 1 of 8:9c1e

SCREEN 45

All three numbers back from your own paper, box by box. The wizard combines them and the twelve words must come back exactly.

You type 9c1e and press enter.

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BUILD THE VAULT - ROUND-TRIP RESULT

1 critic	2 next	3 cloud	4 athlete
5 balcony	6 timber	7 gasp	8 draft
9 paddle	10 silk	11 doll	12 glass

MATCHES the seed you started with.

And they match the paper seed in front of you, word for word? [y/n]y

SCREEN 46

The twelve words come back exactly. The paper is right.

You type y and press enter.

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BUILD THE VAULT - WRITE THE RED STICK

Plug in ONE stick: the one that will carry the NUMBERS 1 AND 2.
Unplug every other USB stick first.

This is the rule that makes the whole scheme safe: with one stick in the machine, two parts cannot land on the same one.

[1] it is plugged in, and it is the only one
[2] skip the USB copy of this part

choose:1

SCREEN 47

One stick at a time, with every other stick unplugged, so both halves can never land on the same one. The half, the plate files and the recovery tools go on, then it is read back.

You type 1 and press enter.

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BUILD THE VAULT - THE RED STICK IS WRITTEN

Written and verified.

Every file was read back and matched its checksum, and the part was re-derived from the stick to confirm the check groups.

Now, before you do anything else:

Write R on this stick in permanent marker.

R = numbers 1 and 2. It is safe to label: a part on its own is nothing.

Then unplug it.

press enter when it is labelled and unplugged

SCREEN 48

Read back and verified. Label it R or B in marker: a half on its own is nothing, so the label is safe. Then unplug it before the other one goes in.

You press enter.

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BUILD THE VAULT - WRITE THE BLUE STICK

Plug in ONE stick: the one that will carry the NUMBERS 2 AND 3.
Unplug every other USB stick first.

This is the rule that makes the whole scheme safe: with one stick in the machine, two parts cannot land on the same one.

[1] it is plugged in, and it is the only one
[2] skip the USB copy of this part

choose:1

SCREEN 49

One stick at a time, with every other stick unplugged, so both halves can never land on the same one. The half, the plate files and the recovery tools go on, then it is read back.

You type 1 and press enter.

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BUILD THE VAULT - THE BLUE STICK IS WRITTEN

Written and verified.

Every file was read back and matched its checksum, and the part was re-derived from the stick to confirm the check groups.

Now, before you do anything else:

Write B on this stick in permanent marker.

B = numbers 2 and 3. It is safe to label: a part on its own is nothing.

Then unplug it.

press enter when it is labelled and unplugged

SPLITSTEEL · the Pi is a blender · steel is the vault

BUILD THE VAULT - WRITE THE GREEN STICK

Plug in **ONE** stick: the one that will carry the **NUMBERS 3 AND 1**.
Unplug every other USB stick first.

This is the rule that makes the whole scheme safe: with one stick in the machine, two parts cannot land on the same one.

[1] it is plugged in, and it is the only one

[2] skip the USB copy of this part

choose:1

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BUILD THE VAULT - THE GREEN STICK IS WRITTEN

Written and verified.

Every file was read back and matched its checksum, and the part was re-derived from the stick to confirm the check groups.

Now, before you do anything else:

Write G on this stick in permanent marker.

G = numbers 3 and 1. It is safe to label: a part on its own is nothing.

Then unplug it.

press enter when it is labelled and unplugged

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BUILD THE VAULT - THE SITTING IS DONE

On the table you now have three numbers, each with a check group, paired into three parts: **RED** 1+2, **BLUE** 2+3, **GREEN** 3+1. In order:

OPEN START-HERE.html ON EACH STICK.

It is a page that opens with a double-click on any computer and walks that part through to finished: burning the disc, making the plates, and where to put it. You do not have to remember any of what follows - it is all on that page, per part.

THE THREE STICKS MUST NEVER MEET.

Take **R**, **B** and **G** each to a different computer, and keep it that way. One stick alone is nothing. Any two on one computer is your whole seed on that computer.

- 1 punch or engrave the keeper plates: two per part, one per number
- 2 make the spare set on your chosen medium
- 3 run RECOVER on this wizard reading only off two finished parts
- 4 only then destroy the old paper seed
- 5 three places: no hiding place holds two different parts
- 6 wipe or destroy this SD, power off, post the heirs' letter

When you are finished with this machine, choose **p** at the menu to power off. Do not pull the plug.
press enter to clear the screen

SCREEN 50

Read back and verified. Label it R or B in marker: a half on its own is nothing, so the label is safe. Then unplug it before the other one goes in.

You press enter.

SCREEN 51

One stick at a time, with every other stick unplugged, so both halves can never land on the same one. The half, the plate files and the recovery tools go on, then it is read back.

You type **1** and press enter.

SCREEN 52

Read back and verified. Label it R or B in marker: a half on its own is nothing, so the label is safe. Then unplug it before the other one goes in.

You press enter.

SCREEN 53

Three parts: two plates each, one stick each, one disc each, and the rule that no hiding place ever holds two different parts. Any two of the three recover.

You press enter.

EPISODE 08

Recover

The bad day: two objects, both halves typed box by box, the words back, and one question before you finish.

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BEFORE ANYTHING

This machine must be a blender: no network lead, no Wi-Fi hardware, no camera that can see this screen, no phone on the table.

Nothing you type is ever written to disk. Power off ends everything.
Is the room clean and the machine air-gapped? [y/n]y

SPLITSTEEL · the Pi is a blender · steel is the vault

RECOVER - FROM THE OBJECTS

One pad copy and one ciphertext copy, or any two parts of three, in a private room, on a blender.

Look at the objects. A two-halves backup carries ONE number on each, marked RED or BLUE. A three-parts backup carries TWO numbers on each, marked RED, BLUE or GREEN, and any two of the three parts will do.

[1] one number on each object - two halves

[2] two numbers on each object - three parts, any two

choose [enter = 1]:1

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RECOVER - TWO HALVES

RED marks the pad and BLUE the ciphertext, on a stick or a disc alike. Plates carry no mark, and it does not matter: the arithmetic comes out the same either way round, so enter two plates in any order.

Two objects, one number each: the RED pad first, then the BLUE ciphertext.

Before the first number goes in:

if this is not a blender, unplug the network lead and switch Wi-Fi off (aeroplane mode) now. The whole seed is in this machine's memory the moment the second object is typed, and nothing must be listening.

press enter when it is off the network - the screen will then clear

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READ IT BACK

the RED PAD, read from the object in front of you

Four characters per box, in the order they appear on the object. Move about freely - nothing you have typed is lost by moving.

Type a command or the characters, then press enter:

1 to 9 go straight to that box
b back one box, leaving it as it is
enter keep this box and move on
q give up on this entry

1 2 3 4

5 6 7 8

check

box 1 of 8:2d94

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RECOVER - THE WORDS

1 critic	2 next	3 cloud	4 athlete
5 balcony	6 timber	7 gasp	8 draft
9 paddle	10 silk	11 doll	12 glass

Write them down before you go on - the next screen clears this one.
press enter to continue - the screen will then clear

SCREEN 54

The air-gap question. The Pi has no radios and no network lead, but the room matters too: no camera on the screen, no phone on the table.

You type y and press enter.

SCREEN 55

The bad day. One pad copy and one ciphertext copy, on a blender. RED marks the pad and BLUE the ciphertext on a stick or a disc; plates carry no mark, and the order does not matter.

You type 1 and press enter.

SCREEN 56

Two objects, one number each, the RED pad then the BLUE ciphertext, and before the first number goes in, a reminder: on anything but a blender, take the machine off the network now.

You press enter.

SCREEN 57

Each half goes in box by box from the paper. The check box at the end catches a misread character and says which column of the paper to look at.

You type 2d94 and press enter.

SCREEN 58

The words are back. Write them down before the screen clears.

You press enter.

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RECOVER SUCCEEDED - ONE QUESTION BEFORE YOU FINISH

You have your words back - the two objects rebuilt the wallet.

Before you do anything with them, this program needs to know one thing, because the safe next step is completely different depending on the answer: **what kind of machine did you just do this on?**

[1] a dedicated blender - never been online, will be wiped
(a kit Pi like this one)

[2] an everyday computer, or I am not sure
(a laptop or PC that has been on the internet)

choose 1 or 2:1

SPLITSTEEL · the Pi is a blender · steel is the vault

RECOVER - THE SEED IS STILL SAFE

Good - nothing untrusted has seen it.

The only machine that has seen both objects is this one, and it goes no further.

- 1 Write the words on a scrap of paper.
- 2 Put them into a wallet on a device that has NOT been on the internet - a hardware wallet (Trezor, Ledger and the like, bought new) is safest. **Never a website, never a wallet on a phone or PC that browses the web.** Confirm the balance is what you expected, then move the funds to a wallet you control. New to this? Choose [?] help at the menu first.
- 3 Burn the scrap.
- 4 Wipe or destroy this SD card.
- 5 Put the two objects back in different buildings.

press enter to clear the screen

SCREEN 59

One question before you finish, because the safe next step depends entirely on what kind of machine just saw both halves.

You type **1** and press enter.

SCREEN 60

On a dedicated blender the seed is still private: into a hardware wallet, never a website, confirm the balance, move the funds, burn the scrap, wipe the card.

You press enter.

EPISODE 09

Help for heirs

Written for someone who has never done any of this. The words are the money.

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HELP - START HERE IF THIS IS NEW TO YOU

If you have inherited this, or just do not know your way around crypto, that is completely normal. Read this once - it is short, and it is the difference between keeping the money and losing it.

The words this machine gives you **ARE** the money.

Like cash, or a key to a safe: whoever has them owns everything inside, at once, with no way to get it back. So:

THE RULES THAT KEEP YOU SAFE

- **Never type the words into a website.** No real service needs them.
- **Never send them to anyone** - not by email, message, or photo.
- Never type them because someone helping you over the phone or chat told you to. That is the commonest way people are robbed.
- A real helper **never needs to take the words away** or keep a copy. They can guide you while YOU hold the paper.
- In any doubt, **stop**. The money is safe as long as the words are. Nothing about this is urgent, whatever anyone tells you.

press enter for the next page

SCREEN 61

The heir's first screen. The words are the money; never type them into a website, never send them to anyone, and nobody real ever needs to take them away.

You press enter.

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HELP - WHAT TO DO WITH THE WORDS

To reach the money, the words go into **wallet software** on a device that has not been on the internet - a **hardware wallet** (a small dedicated device such as a Trezor or Ledger, bought new and sealed) is the safest choice. Never put them into a wallet on an everyday phone or computer that browses the web.

Once the wallet shows the balance, move it to a wallet that is yours, then treat these words as spent.

IF YOU NEED A PERSON TO HELP

- Best of all: someone you already know and would trust with cash.
- A solicitor experienced in digital assets.
- A regulated crypto custody firm - but **you** approach them; never accept an offer that came to you, and they still never need the words themselves.
- Hardware-wallet makers give free setup guides and support. Real support helps you use the device and **never asks for your words**. If "support" asks for the words, it is a thief. Walk away.

The full manual, on the disc and the memory stick, says all of this again with more detail, including how to do it with no computer. press enter to continue - the screen will then clear

SCREEN 62

What to do with the words: a hardware wallet bought new, then move the money and treat these words as spent. Who can safely help, and the one sign of a thief.

You press enter.

Two sticks, two numbers on paper. From here on, one half at a time.

The blender is the only machine that ever holds your whole seed. Everything after it handles one half, and a half is a random number, so the computers that make the copies need no trust and no skill. One rule replaces all of that: **the two halves never meet on the same computer.**

- **The sticks.** Each carries one half, its plate files, the recovery tools and START-HERE.html, a page that opens with a double-click and walks that half through to finished. Take R to one computer and B to a different one.
- **The discs.** One M-DISC per half, burned on that half's computer from the files on its stick.
- **The steel.** A keeper plate and a spare for each half, punched with the jig from the plate files, eight groups and the check group. No labels, ever: the plates carry no red or blue and no name, because the arithmetic does not care which is which.
- **Then prove it, then split it.** Run RECOVER on the kit reading only off the finished objects. Only then destroy the old paper seed. Two halves, three copies each, and no hiding place holds both halves. Wipe the SD card, power off, post the heirs' letter.

INHERITANCE

Split the secret, not the trust.

Copies of the same half never add up. Twenty people holding twenty copies of the ciphertext hold nothing between them, which is what makes a split different from a multisig: nobody you give a half to can ever spend, alone or together. The whole design of the inheritance is deciding who could hold one half and also reach the other.

- **Many ciphers.** Spread the ciphertext as widely as is useful: a solicitor, a family member, a hole in the garden, even a dead man's switch in an online account. A hacked account learns a random string and nothing else.
- **Few pads.** Keep the pad copies few, in places only you control, never online, never with anyone who holds a cipher. Once a thief has a pad, every cipher copy in the world is a place they can finish the job.
- **The one rule.** No person, account or place ever holds one half and the whereabouts of the other. A solicitor with a cipher and a sealed letter saying where a pad is holds the whole secret.
- **The after-death path takes two.** The will names where the pad is, and whoever holds the will holds no cipher. A cipher sits with someone who does not know where the pad is. Either alone has nothing. Together, after you, they recover it.

THE BAD DAY

For the person who inherits this.

The wizard has a help screen written for you, and the recovery is two grids typed from two objects. The words it gives you are the money. Nothing about this is urgent, whatever anyone says.

- Read the help screens first: choose [?] at the menu.
- Gather one pad object and one ciphertext object: a red stick or disc and a blue one, or two plates in any order.
- Sit at the kit in a private room, choose [2] recover, and type each half box by box. The check box catches a misread character and says which column of the object to look at.
- Write the words on paper when the screen shows them, before pressing enter.
- Put them into a hardware wallet bought new, never a website, never a phone or PC that browses the web. Confirm the balance, then move everything to a wallet that is yours.
- Burn the scrap, wipe the SD card, and put the two objects back in different buildings until the old wallet is empty.
- Real help never needs your words. If anyone offering support asks for them, they are a thief. Walk away.

This quick start was generated from program build **0396da41**. The field manual, with the three printed cards, is splitsteel-field-manual.pdf beside this file.